

**Let there be light**

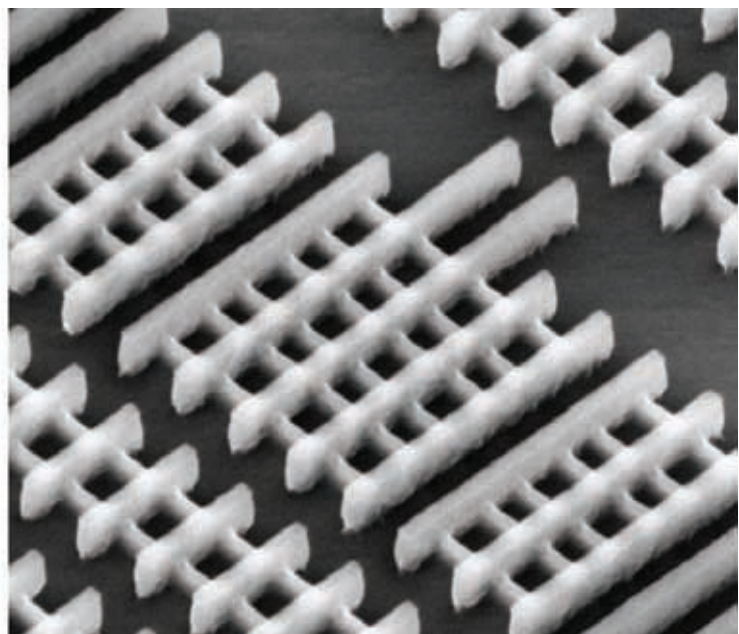
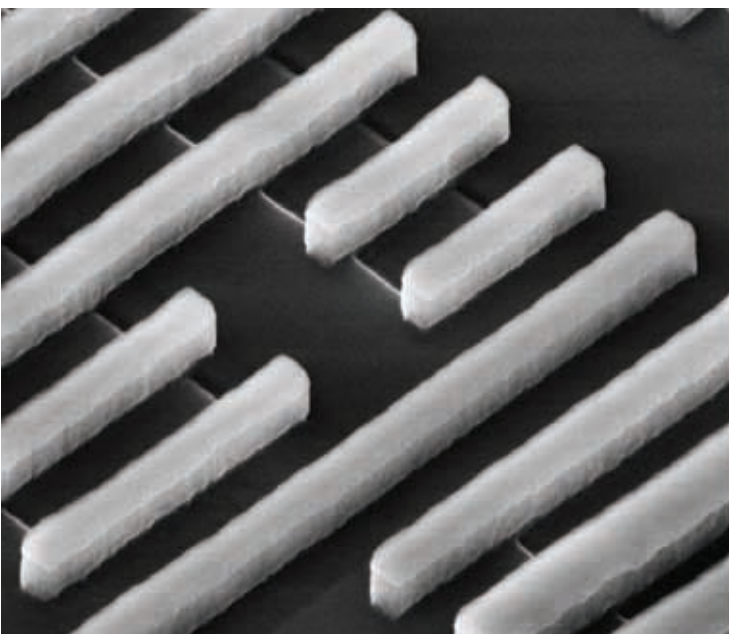
A 22nm transistor can switch on/off over a 100 billion times per sec. You would need 2,000 years to flick a light switch on and off that many times

**Teeny tiny**

A 22nm tri-gate transistor's gates are so small, you could fit more than 4,000 of them across the width of a human hair

# Two's company and three's Moore!

At first glance it might seem Intel has hopped aboard the increasingly popular 3D bandwagon, but on closer scrutiny this isn't just a buzzword to encourage hype. We take a look at yet another 3D technology...this one happily far away from the realm of displays.



Michael Browne  
[michael.browne@thinkdigit.com](mailto:michael.browne@thinkdigit.com)

In the beginning of May, Intel made public what is now being believed to be the most significant advances in microprocessor technology, and what industry old timers are claiming “reinvents the semiconductor”. While we’re pretty familiar with PR agencies’ overstating of facts, what we media folk term politely as “purple prose”, the concept as unveiled by Intel seems to hold more than a passable amount of water.

The fact is this is no secret. It has been nearly 10 years since rumours first surfaced about Intel working on some revolutionary sort of transistor, and till

then, small, seemingly inconspicuous news tidbits emerged, but these were rather insignificant to the technological advances that were being unveiled over the years. After all, seeing-is-believing, and there was nothing to see. Until now that is. Some have also joked that this development was to ensure their beloved Gordon E. Moore’s famous words (aka Moore’s Law) held weight just a bit longer.

In fact, Mr. Moore himself had something more to say on the occasion of the 3D transistor announcement – “For years we have seen limits to how small transistors can get. This change in the basic structure is a truly revolutionary approach, and one that should allow Moore’s Law, and the historic pace of innovation, to continue”

The first processors to utilise these 3D or Tri-Gate transistors in lieu of the traditional planar (2D) transistors will be Ivy Bridge, the codename for Intel’s upcoming 22nm processor family, due for market release sometime in 2012. This announcement is veritably huge, because it is not simply a die shrink, or the presence of a proprietary addition like Quick Sync, or Fast Caching, but represents a complete change of the building block of any integrated circuit – the transistor.

## TRANSISTORS: the traditional way

Since the birth of the transistor in 1947, and William Shockley’s bipolar junction transistor three years later, the humble